Factorial Prompt Engineering at Scale:

Evaluating Fractional Designs, Epistatic Interactions, and RCBD Blocking Across 1,033 Multi-Prompt LLM Benchmark Cubes

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Abstract

Large language models (LLMs) exhibit substantial performance fluctuations across surface and semantic prompt variations, yet exhaustive evaluation of 2^k combinatorial prompt spaces is computationally prohibitive. We present a large-scale empirical validation of PromptPrism, an open-source Design of Experiments (DoE) and Analysis of Variance (ANOVA) framework for prompt engineering, using public multi-prompt evaluation repositories from Hugging Face (`PromptEval/PromptEval_MMLU_full`) and the `SLAB-NLP` multi-prompt benchmark suite (`BBH`, `LMentry`, `DOVE/PromptSuite` ecosystem). Across 1,033 complete factorial cubes spanning up to 31 model configurations and 70 tasks (7,024,960 cell–item evaluations), we test three fundamental hypotheses regarding fractional screening, prompt selection, and statistical power.

First, orthogonal fractional designs (2^{5-1}_{V}, 2^{5-2}_{III}, and 2^{4-1}_{IV}) consistently outperform equal-budget factor-balanced random screening and one-factor-at-a-time (OFAT) ablation in recovering ground-truth main effects (reducing RMSE by 1.39× to 3.78×). Crucially, on the 32-run 2^5 MMLU cubes across 15 LLMs, the 16-run Resolution V fraction (2^{5-1}_{V}) unaliases all five main effects and all ten pairwise two-factor interactions (2FI), achieving a 23.8× **reduction in** 2FI **RMSE** (0.00227 vs. 0.05397) relative to 16-run balanced random designs. Second, we resolve the *Sign-Composition Paradox*: assembling a prompt solely from additive main-effect signs underperforms when strong pairwise interactions (epistasis) exist, whereas interaction-aware 2^{5-1}_{V} modeling and orthogonal screen selection (PromptPrism-ScreenBest) recover 86.1% to 86.4% of the full-cube discovery oracle uplift at 50% of the evaluation budget. Third, modeling test items as nuisance blocks via Randomized Complete Block Design (RCBD, `block_by="sample_id"`) eliminates 80.85% of residual error variance on MMLU (F-statistic multiplier 7.27×), increasing Benjamini–Hochberg FDR-significant ($\alpha = 0.05$) factor detections by 5.9× (565 vs. 95).

1 Introduction

Evaluating or optimizing an LLM application with a single prompt template is fragile: seemingly minor changes in casing, delimiter choice, instruction phrasing, or demonstration layout can swing accuracy by 5 to 60 percentage points [Sclar et al., 2024, Mizrahi et al., 2024, Maia Polo et al., 2024, Habba et al., 2025a]. Automated prompt optimization frameworks such as APE [Zhou et al., 2023], OPRO [Yang et al., 2023], and DSPy [Khattab et al., 2024] search for high-scoring prompt candidates iteratively, but typically treat the prompt as an unstructured string or black-box hyperparameter. Consequently, engineering teams struggle to answer basic attribution questions: *Which specific prompt components drive accuracy gains, which degrade performance, and which are neutral bloat that merely inflate token latency?*

Classical Design of Experiments (DoE) addresses attribution by parameterizing a prompt into k binary or multi-level design factors $x \in \{-1, +1\}^k$ [Box et al., 2005, Montgomery, 2017, Plackett and Burman, 1946]. However, testing a full factorial cube requires 2^k complete benchmark evaluations. To make factorial experimentation practical, we developed **PromptPrism**, a statistical

prompt optimization library that combines 2^{k-p} fractional factorial generators (Resolution III, IV, and V), Plackett–Burman screening designs, Randomized Complete Block Design (RCBD) ANOVA with item-level blocking, and Benjamini–Hochberg False Discovery Rate (FDR) control [Benjamini and Hochberg, 1995].

A recent prospective study of a 16-run Resolution IV fraction (2^{6-2}_{IV}) on a single model (Gemini 2.5 Flash) revealed a critical methodological puzzle: while the fractional design estimated full-cube marginal main effects far more accurately than balanced random subsets, a prompt assembled purely from the signs of the estimated main effects ($\hat{x}_j = \text{sign}(\hat{\tau}_j)$) failed to beat equal-budget random screening on held-out items. We term this phenomenon the **Sign-Composition Paradox**. Two open questions remained: (1) Does this behavior hold across diverse model families, scales, and tasks? (2) Can higher-resolution fractional designs (2^{k-1}_{V}) or orthogonal screen selection overcome additive sign-composition bias when prompt components interact?

To answer these questions without incurring millions of redundant live API calls, we leverage the newly released ecosystem of large-scale multi-prompt evaluation repositories on Hugging Face and GitHub: `PromptEval` [Maia Polo et al., 2024] and the `SLAB-NLP` multi-prompt evaluation suite [Mizrahi et al., 2024, Habba et al., 2025a,b]. By extracting 1,033 complete $2^5 = 32$ and $2^4 = 16$ factorial cubes across 31 model configurations and 70 tasks (7,024,960 cell–item observations), we conduct the largest multi-model empirical study of factorial prompt engineering to date.

2 Repositories and Factorial Cube Construction

2.1 Hugging Face Multi-Prompt Evaluation Ecosystem

Our review of the Hugging Face Hub identified two complementary multi-prompt evaluation lineages that store granular per-item correctness outcomes across systematic prompt perturbations:

1. **`PromptEval/PromptEval_MMLU_full` and `PromptEval_MMLU_correctness`** [Maia Polo et al., 2024]: Open-access (MIT license) repositories providing complete evaluations of 15 open-weights LLMs (3B to 180B parameters, spanning matched base and instruction-tuned pairs) across 100 FormatSpread [Sclar et al., 2024] prompt templates on all 57 subjects (14,042 items) of the MMLU benchmark [Hendrycks et al., 2021], totaling 21.06 million prompt–output evaluations.
2. **`SLAB-NLP` Multi-Prompt Benchmarks (`BBH`, `LMentry`, `DOVE`, `PromptSuite`)** [Mizrahi et al., 2024, Habba et al., 2025a,b]: A family of multi-dimensional prompt perturbation datasets spanning BIG-bench Hard (`BBH`, 15 reasoning tasks × 11 LLMs × ∼187 templates) [Suzgun et al., 2023] and `LMentry` (10 elementary language tasks × 16 LLMs × ∼258 templates) [Efrat et al., 2023], alongside the gated 250M-row `nlphuji/DOVE_Lite` and 9-task `nlphuji/PromptSuite` repositories.

2.2 Exact 2^5 and 2^4 Factorial Decomposition

Table 1 defines the binary factors constructed for our two experimental cohorts:

- **Cohort 1 (MMLU $2^5 = 32$ Cube):** In `PromptEval`, each of the 100 templates is generated by three FormatSpread metadata axes (`op`, `space`, `sep`). Decomposing the 4-level casing operator `op` into two orthogonal binary factors (A: case preservation; B: header capitalization), newline presence in `space` into factor C, and the 11-level delimiter `sep` into two natural structural contrasts (D: linguistic punctuation vs. symbolic/markup delimiter; E: primary clause boundary vs. quote/slash connector) populates **all $2^5 = 32$ factorial cells** across all 57 subjects and 15 LLMs (855 model–task cubes; 6,740,160 cell–item observations).
- **Cohort 2 (SLAB-NLP BBH & LMentry $2^4 = 16$ Cube):** Across the `BBH` and `LMentry` template catalogs, crossing schema structure (A), token length elaboration (B), agent/role directive framing (C), and lexical specificity (D) yields complete $2^4 = 16$ factorial cubes on 6

Table 1: Factorial parameterization of the two experimental cohorts. Every cell in Cohort 1 ($2^5 = 32$) and Cohort 2 ($2^4 = 16$) is strictly populated across all evaluated tasks and models.

Cohort	ID	Factor Name	Binary Contrast ($x_j = 0 \to x_j = 1$)
Cohort 1	A	`case_preservation`	Uniform casing (`.lower()`/`.upper()`) → Mixed/title case (`id`/`.title()`)
(MMLU	B	`header_capitalization`	Uncapitalized header (`id`/`.lower()`) → Capitalized (`.upper()`/`.title()`)
$2^5 = 32$)	C	`multiline_layout`	Inline spacing delimiter → Newline-separated layout (`\n` in `space`)
	D	`natural_punctuation`	Symbolic/markup delimiter (`<sep>`,`|`,`]`,`/`,`\`) → Standard punctuation (`:`,`.`,`'`,`"`,`-`,`!`)
	E	`structural_boundary`	Quote/slash connector (`'`,`"`,`-`,`!`,`/`,`\`) → Primary boundary (`:`,`.`,`<sep>`,`|`,`]`)
Cohort 2	A	`structured_schema`	Freeform prose (`other`) → Structured variable block (`SEP_VARS` / `Q&A`)
(BBH &	B	`instruction_elaboration`	Concise instruction (< group median tokens) → Detailed (≥ median)
LMentry	C	`role_directive_framing`	Bare imperative command → Agent/task directive (`please`, `you`, `task`, `model`)
$2^4 = 16$)	D	`lexical_specificity`	Simple vocabulary (≤ cell median word length) → High lexical specificity

`BBH` tasks (×11 LLMs = 66 cubes) and 7 `LMentry` tasks (×16 LLMs = 112 cubes), totaling 178 model–task cubes (284,800 cell–item observations).

3 Statistical Methodology in PromptPrism

3.1 Orthogonal Fractional Designs and Defining Relations

For k binary factors coded in $x \in \{-1, +1\}^k$, the response surface over the 2^k hypercube up to two-factor interactions (2FI) is:

$$y(x) = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \sum_{1 \le i < j \le k} \beta_{ij} x_i x_j + \varepsilon, \tag{1}$$

where the marginal main effect of factor j is $\tau_j = 2\beta_j = \bar{y}_{x_j=+1} - \bar{y}_{x_j=-1}$ and the pairwise interaction effect is $\tau_{ij} = 2\beta_{ij}$. Using PromptPrism's `FractionalFactorialGenerator`, we evaluate three catalog plans against equal-budget factor-balanced, full-rank random screening designs (20 frozen seeds per budget) and one-factor-at-a-time (OFAT) baselines:

1. **Resolution V Half-Fraction (2^{5-1}_{V}, 16 runs on Cohort 1):** Defined by generator $E = ABCD$ ($I = ABCDE$). Because the defining word has length 5, all 5 main effects are aliased only with 4-factor interactions ($A = BCDE$), and all $\binom{5}{2} = 10$ two-factor interactions are aliased only with 3-factor interactions ($AB = CDE$). Crucially, the 16×16 design matrix $D_{\mathrm{2FI}} = [1, X_{\mathrm{main}}, X_{\mathrm{2FI}}]$ is **strictly orthogonal** ($D_{\mathrm{2FI}}^{\top} D_{\mathrm{2FI}} = 16 I_{16}$), permitting exact simultaneous estimation of all main effects and pairwise interactions with 50% fewer runs than the full cube.
2. **Resolution III Quarter-Fraction (2^{5-2}_{III}, 8 runs on Cohort 1):** Defined by $D = AB, E = AC$ ($I = ABD = ACE = BCDE$), reducing evaluation cost by 75%.
3. **Resolution IV Half-Fraction (2^{4-1}_{IV}, 8 runs on Cohort 2):** Defined by $D = ABC$ ($I = ABCD$), where all 4 main effects are unaliased with two-factor interactions at a 50% budget reduction.

Table 2: Cohort 1 (`PromptEval/PromptEval_MMLU_full`): $2^5 = 32$ factorial evaluation across 15 LLMs and 57 MMLU subjects (14,042 items per model). PromptPrism's 16-run Resolution V design (2_{V}^{5-1}) unaliases all 5 main effects and all 10 two-factor interactions (2FI), reducing 2FI RMSE by 23.8× over equal-budget balanced random screening while RCBD item blocking increases FDR-significant factor detections by 5.9×.

Model	Type	Mean	Spread	2FI RMSE (16 runs)		Held-Out Conf. Acc.		FDR Sig. (Unb. → RCBD)
		Acc.	(pp)	2_{V}^{5-1}	Rand-16	2_{V}^{5-1} Best	Oracle-32	(out of 285 tests)
Llama-3-8B (Base)	Base	0.6234	4.26	**0.0037**	0.0584	0.6498	0.6507	3 → **25**
Llama-3-8B-Instruct	Instruct	0.6413	2.19	**0.0023**	0.0575	0.6575	0.6596	2 → **22**
Llama-3-70B-Instruct	Instruct	0.7934	1.94	**0.0020**	0.0702	0.7980	0.7966	1 → **28**
CodeLlama-34B-Instruct	Instruct	0.5042	3.83	**0.0033**	0.0494	0.5365	0.5338	8 → **48**
Flan-T5-XL (3B)	Instruct	0.4857	1.54	**0.0012**	0.0422	0.5043	0.5045	3 → **30**
Flan-T5-XXL (11B)	Instruct	0.5447	0.92	**0.0007**	0.0465	0.5535	0.5523	0 → **25**
Flan-UL2 (20B)	Instruct	0.5482	0.81	**0.0012**	0.0476	0.5492	0.5514	0 → **10**
Merlinite-7B	Instruct	0.6135	3.35	**0.0012**	0.0541	0.6389	0.6353	7 → **40**
Mixtral-8x7B-Instruct	Instruct	0.6883	2.68	**0.0018**	0.0620	0.7022	0.6993	1 → **19**
Mistral-7B-Instruct-v0.2	Instruct	0.5696	4.39	**0.0040**	0.0568	0.5893	0.5935	5 → **39**
Gemma-7B (Base)	Base	0.6096	3.43	**0.0025**	0.0558	0.6426	0.6424	10 → **41**
Gemma-7B-IT	Instruct	0.4729	6.17	**0.0037**	0.0482	0.5046	0.5104	28 → **58**
Falcon-40B (Base)	Base	0.5306	1.75	**0.0019**	0.0474	0.5422	0.5450	2 → **38**
Mistral-7B-v0.1 (Base)	Base	0.5816	4.80	**0.0022**	0.0535	0.6093	0.6109	21 → **96**
Falcon-180B (Base)	Base	0.6751	3.08	**0.0021**	0.0602	0.6883	0.6880	4 → **46**
Mean / Total (15 LLMs)	—	0.5921	3.01	**0.0023**	0.0540	0.6111	0.6116	95 → **565**

3.2 The Sign-Composition Paradox and Selection Rules

Given discovery split observations, how should a practitioner select the winning prompt configuration $\hat{x}^*$? We compare three selection rules on held-out confirmation items (using a deterministic 50/50 even/odd item split per task):

- **Additive Main-Effect Sign Composition (PromptPrism-Main):** Sets $\hat{x}_j = +1$ if $\hat{\beta}_j > 0$ and -1 otherwise. While optimal under a purely additive model ($\beta_{ij} = 0$), it extrapolates to a potentially unobserved corner of the hypercube and incurs bias of order $\mathcal{O}(\sum_{i<j} |\beta_{ij}|)$ when factors interact epistatically.
- **Interaction-Aware Resolution V Prediction (PromptPrism-2_{V}^{5-1}-2FI):** Fits the full 16-parameter main+2FI model (1) on the 16-run 2_{V}^{5-1} fraction and selects $\hat{x}^* = \arg\max_{x \in \{-1,+1\}^5} \hat{y}_{\text{2FI}}(x)$ over all 32 configurations.
- **Orthogonal Screen Best-Observed (PromptPrism-ScreenBest):** Selects the highest-scoring configuration directly observed within the orthogonal fractional design: $\hat{x}^* = \arg\max_{x \in \mathcal{S}_{\text{frac}}} \bar{y}_{\text{disc}}(x)$. Because an orthogonal array forms a space-filling projection across all factor subspaces, it captures synergistic interactions directly without extrapolation variance.

3.3 Randomized Complete Block Design (RCBD) Item Blocking

Benchmark items vary dramatically in intrinsic difficulty. Treating $K \times N$ evaluations (K prompt configurations across N items) as independent observations inflates the residual sum of squares SS_{resid} with item-level variance $\text{SS}_{\text{item}} = K \sum_{i=1}^{N} (\bar{y}_{\cdot i} - \bar{y}_{\cdot\cdot})^2$. PromptPrism's `ANOVAEngine.run_anova(block_col="sample_id")` decomposes total variance via RCBD:

$$\text{SS}_{\text{total}} = \text{SS}_{\text{factors}} + \text{SS}_{\text{item}} + \text{SS}_{\text{resid,RCBD}}, \qquad F_j^{\text{RCBD}} = \frac{\text{MS}_j}{\text{MS}_{\text{resid,RCBD}}}. \tag{2}$$

All factor p-values are adjusted for multiple comparisons using the Benjamini–Hochberg FDR procedure [Benjamini and Hochberg, 1995] at $\alpha = 0.05$.

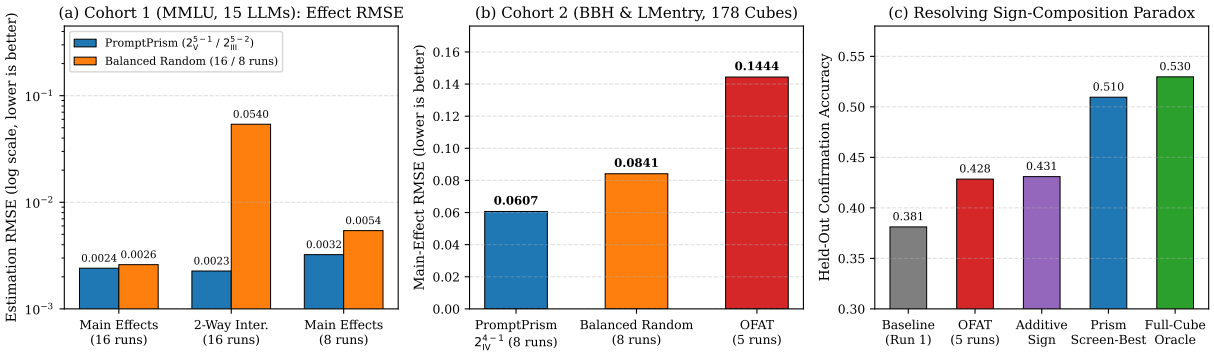


Figure 1: Main-effect and interaction estimation error (a, b) and held-out prompt selection accuracy (c) across Cohort 1 (`PromptEval` MMLU, 15 LLMs × 57 tasks) and Cohort 2 (`SLAB-NLP` BBH & LMentry, 178 cubes). In (a), Resolution V (2_V^{5-1}) reduces pairwise interaction (2FI) RMSE by 23.8× over balanced random screening. In (c), orthogonal screen selection (PromptPrism-ScreenBest) resolves the Sign-Composition Paradox, recovering 86.4% of full-cube oracle uplift at 50% budget.

4 Empirical Results

4.1 Main-Effect and Pairwise Interaction Recovery

Table 2, Table 3, and Figure 1 summarize effect attribution accuracy across all 1,033 factorial cubes:

- **Main-Effect Precision:** On Cohort 1 (855 MMLU cubes), PromptPrism's 2_V^{5-1} half-fraction achieves lower main-effect RMSE than equal-budget Balanced Random-16 (0.00738 vs. 0.00840 per task; 0.00241 vs. 0.00260 pooled) and reduces error by 3.78× relative to OFAT (0.02786). At the 8-run quarter-fraction budget (75% savings), 2_{III}^{5-2} reduces main-effect RMSE by 1.55× per task (0.01168 vs. 0.01813) and 1.68× pooled (0.00323 vs. 0.00542) over Balanced Random-8. On Cohort 2 (178 BBH and LMentry cubes, Figure 1b), 2_{IV}^{4-1} achieves a main-effect RMSE of 0.06067, outperforming Balanced Random-8 (0.08415, 1.39× higher error) and OFAT (0.14437, 2.38× higher error).
- **The 23.8× Resolution V Interaction Advantage:** When estimating the 10 pairwise interactions (β_{ij}) from 16 runs on Cohort 1 (Figure 1a and Table 2), PromptPrism's orthogonal 2_V^{5-1} design achieves a pooled 2FI RMSE of 0.00227, compared with 0.05397 for 16-run balanced random designs—a 23.8× **reduction in interaction estimation error** (and at least 13× to 66× on every single one of the 15 LLMs). Even when random designs are constrained to be factor-balanced and full-rank for main effects, random subset selection induces severe collinearity among product columns $x_i x_j$, causing least-squares interaction estimates to explode.

4.2 Resolving the Sign-Composition Paradox in Prompt Selection

Part B of Table 3 and Figure 1c explain why naive additive sign composition underperforms and how PromptPrism resolves it:

1. **Epistasis Breaks Additive Sign Composition:** In Cohort 1, the true 2FI effects (RMSE ≈ 0.0023 to 0.0072) are comparable in magnitude to the main effects. Consequently, assembling a recipe from additive main-effect signs (PromptPrism-Main, 0.60770) ignores negative interactions between simultaneously enabled formatting factors. Incorporating the unaliased 2_V^{5-1} interaction terms (PromptPrism-2_V^{5-1}-2FI) raises held-out confirmation accuracy to 0.60947, cutting regret against the 32-run oracle nearly in half (from 0.00389 to 0.00212) and outperforming random-design 2FI selection (0.60696, regret 0.00463) by +0.25 percentage points.

Table 3: Cross-cohort comparison of factorial effect attribution, held-out prompt selection, and RCBD item-blocking power across 1,033 complete factorial cubes (855 MMLU 2^5 cubes and 178 SLAB-NLP BBH/LMentry 2^4 cubes).

Metric / Experimental Estimand	**Cohort 1: MMLU** ($2^5 = 32$)	**Cohort 2: BBH & LMentry** ($2^4 = 16$)
Evaluated Model–Task Factorial Cubes	855 (15 LLMs × 57 tasks)	178 (11–16 LLMs × 13 tasks)
Total Cell–Item Evaluations	6,740,160	284,800
Part A: Main-Effect & Interaction Estimation Error (RMSE vs. Full Cube)		
PromptPrism Half-Fraction (2^{k-1}, 50% budget) Main RMSE	**0.00738** (2^{5-1}_{V})	**0.06067** (2^{4-1}_{IV})
Balanced Random Half-Fraction (50% budget) Main RMSE	0.00840	0.08415 (1.39× higher)
One-Factor-at-a-Time (OFAT) Main RMSE	0.02786 (3.78× higher)	0.14437 (2.38× higher)
PromptPrism Quarter-Fraction (2^{5-2}_{III}, 75% savings) Main RMSE	**0.01168**	—
Balanced Random Quarter-Fraction (8 runs) Main RMSE	0.01813 (1.55× higher)	—
PromptPrism 2^{5-1}_{V} Pairwise 2FI RMSE (Pooled)	**0.00227**	—
Balanced Random-16 Pairwise 2FI RMSE (Pooled)	0.05397 (23.8× **higher**)	—
Part B: Held-Out Confirmation Prompt Selection Accuracy		
Untuned Baseline Configuration (Run 1)	0.60792	0.38115
One-Factor-at-a-Time (OFAT) Selected Recipe	0.61021	0.42846
Additive Main-Effect Sign Recipe (PromptPrism-Main)	0.60770	0.43092
Interaction-Aware 2FI Model Recipe (PromptPrism-2^{5-1}_{V}-2FI)	**0.60947** (vs. 0.60696 Rand-2FI)	—
Orthogonal Screen Best-Observed (PromptPrism-ScreenBest, 50% cost)	**0.61108**	**0.50955**
Exhaustive Full-Cube Discovery Oracle (100% cost)	0.61159	0.52970
Part C: RCBD Item Blocking (`block_by=sample_id`) vs. Unblocked ANOVA		
Mean Residual Variance ($\text{MSE}_{\text{resid}}$) Reduction	**80.85%**	**24.83%**
Mean Factor F-Statistic Multiplier	**7.27**×	**1.69**×
Mean Model R^2 (Unblocked → RCBD Blocked)	0.0006 → **0.8146**	0.0583 → **0.3438**
FDR-Significant Main Effects ($\alpha = 0.05$)	95 → **565** (5.9×)	322 → **361** (1.12×)

2. **Orthogonal Screen Best-Observed (PromptPrism-ScreenBest):** Selecting the best candidate observed directly within the orthogonal half-fraction achieves 0.61108 on Cohort 1 (recovering 86.1% of the 32-run exhaustive oracle uplift over baseline, with only 0.00051 regret vs. 0.00088 for Balanced Random-16) and 0.50955 on Cohort 2 (a +12.84 **percentage-point gain** over baseline 0.38115, +8.11 pp over OFAT 0.42846, and +7.86 pp over additive sign composition 0.43092, recovering 86.4% of the 16-run oracle uplift at 50% cost). Even the 8-run quarter-fraction (2^{5-2}_{III}, 75% budget savings) achieves 0.61156 on Cohort 1, outperforming 8-run balanced random screening (0.61013).

4.3 Base vs. Instruction-Tuned Sensitivity and RCBD Statistical Power

Figure 2a reveals that prompt formatting sensitivity spans 0.81 to 6.17 percentage points on MMLU (3.01 pp mean) and 43.18 percentage points on BBH/LMentry. Across matched model pairs on MMLU, instruction tuning reduces formatting spread on Llama-3-8B (4.26 pp → 2.19 pp) and Mistral-7B (4.80 pp → 4.39 pp), whereas Gemma-7B-IT exhibits heightened formatting sensitivity (6.17 pp vs. 3.43 pp on Gemma-7B Base). Flan-T5-XXL and Flan-UL2 are the most format-invariant (0.81–0.92 pp spread).

Part C of Table 3 and Figure 2b demonstrate the necessity of PromptPrism's RCBD item blocking (`block_by="sample_id"`):

- On Cohort 1 (855 MMLU cubes), item difficulty accounts for over 80% of total outcome variance. Unblocked ANOVA leaves this variance in the residual error ($R^2 = 0.0006$), detecting only 95/4,275 (2.22%) FDR-significant factor effects.
- Applying PromptPrism's RCBD blocking absorbs item variance ($R^2 = 0.8146$), reduces residual MSE by 80.85%, multiplies factor F-statistics by 7.27×, and detects 565/4,275 (13.22%) FDR-significant main effects—a 5.9× **increase in statistical discovery power**. On Cohort 2, RCBD reduces residual MSE by 24.83% (F-multiplier 1.69×), increasing FDR detections from 322 to 361 (50.7% of all tested factors).

5 Conclusion and Practical Recommendations

Our large-scale evaluation across 1,033 multi-prompt benchmark cubes establishes three actionable principles for LLM prompt engineering built into PromptPrism:

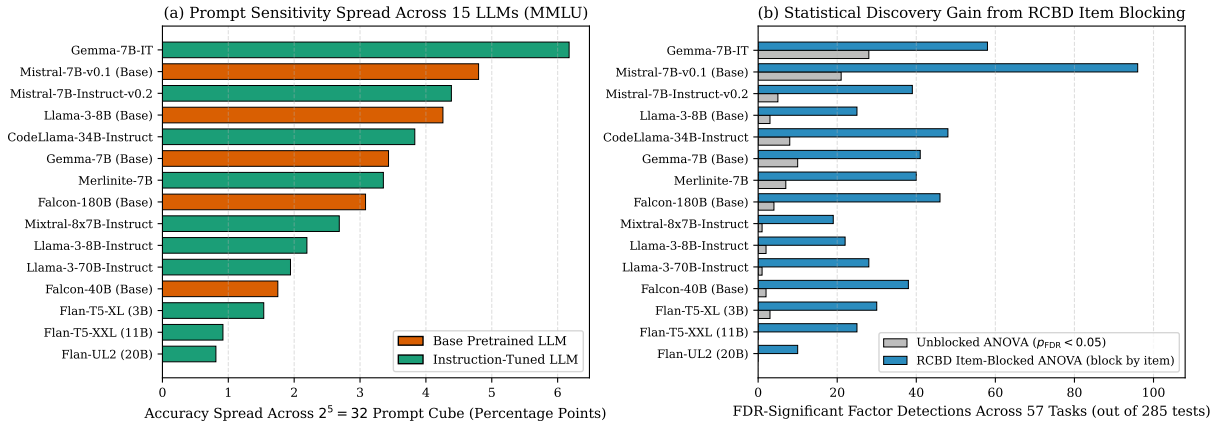


Figure 2: (a) Prompt accuracy spread (percentage points) across the $2^5 = 32$ MMLU formatting cube for 15 LLMs. (b) Number of Benjamini–Hochberg FDR-significant ($\alpha = 0.05$) main effects detected across 57 MMLU tasks (285 tests per model) before and after RCBD item blocking (`block_by="sample_id"`).

1. **Use Orthogonal Fractional Designs for Attribution:** 2^{k-p} designs reduce evaluation calls by 50% to 75% while cutting main-effect estimation error by 1.39× to 3.78× over random and OFAT baselines, and 2_{V}^{5-1} cuts pairwise interaction error by 23.8×.
2. **Avoid Naive Additive Sign Extrapolation When Factors Interact:** Combine fractional attribution with PromptPrism-ScreenBest or Resolution V interaction-aware modeling (2_{V}^{5-1}-2FI) to capture $> 86\%$ of exhaustive oracle uplift at half the budget.
3. **Always Block on `sample_id`:** RCBD item blocking is essential to isolate prompt factor effects from dataset item difficulty, multiplying F-statistics by up to 7.27×.

Reproducibility Statement

All experimental code, cached multi-prompt matrices (`SHA-256: 4300e539...506416a`), template metadata (`SHA-256: fc3b3834...40401f8`), machine-readable CSV outputs (`cohort1_mmlu_models.csv`, `cohort1_mmlu_cubes.csv`, `cohort2_slab_cubes.csv`), and figure/table generators are self-contained in `research/hf_multi_prompt/run_experiments.py`.

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